CS6650 Assignment 1: SkiTrak

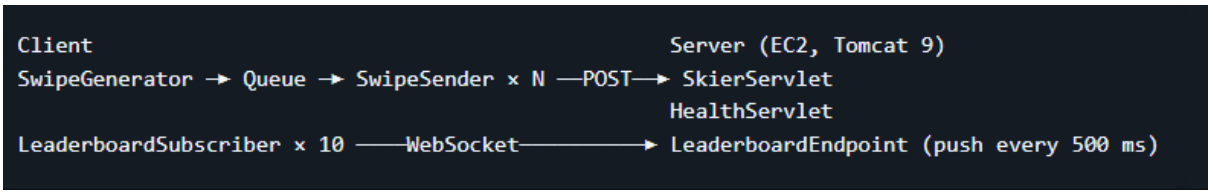
Student Name: Yu-Lun Juan

1. Git Repository

https://github.com/yulunjuan/cs6650-assignment1

2. Design Document

- Architecture



- Major Classes

- SwipeGenerator: one thread, creates all 500K swipes into a LinkedBlockingQueue
- SwipeSender: takes swipes from the queue and sends POSTs, retries up to 5 times with exponential backoff
- LeaderboardSubscriber: one WebSocket per resort, records push latency, reconnects on drop
- Metrics: shared AtomicInteger counters
- RequestRecord, Stats (Part 2): per-request records and statistics

- Threading Model

- One generator thread fills the queue before sending starts, so senders never wait
- Warmup: 32 threads × 1,000 swipes, timed separately with a CountDownLatch
- Main: remaining 468,000 swipes split across 200 threads
- All senders share one HttpClient (connection pool, keep-alive)
- Part 2: each sender stores records in its own list; merged and written to CSV after the run

- WebSocket Strategy

- 10 persistent connections (one per resort), opened before ingest and closed after
- On close or error, reconnect with backoff (up to 5 attempts)
- Part 2 uses a separate HttpClient for WebSockets

- Little's Law

Throughput = threads / RTT

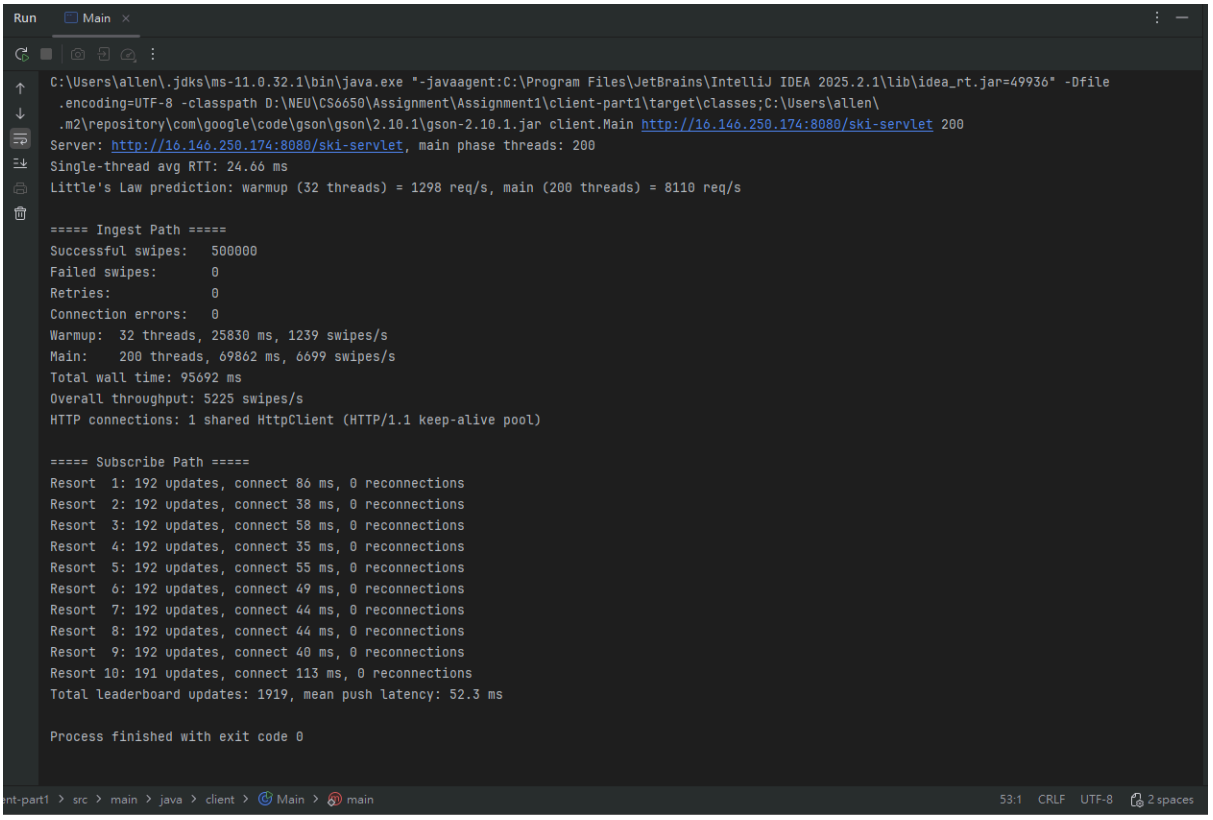
Run	RTT	Phase	Predicted	Measured
Part 1	24.66 ms	Warmup (32)	1,298/s	1,239/s
Part 1	24.66 ms	Main (200)	8,110/s	6,699/s
Part 2	26.37 ms	Warmup (32)	1,213/s	1,178/s
Part 2	26.37 ms	Main (200)	7,584/s	7,570/s

The gap in Part 1's main phase comes from client-side overhead and network variation. Server CPU stayed low during the runs, so the server was not the bottleneck.

Push latency includes clock skew between client and server. My laptop clock was 2.47 s ahead in the first run; results were measured again after syncing.

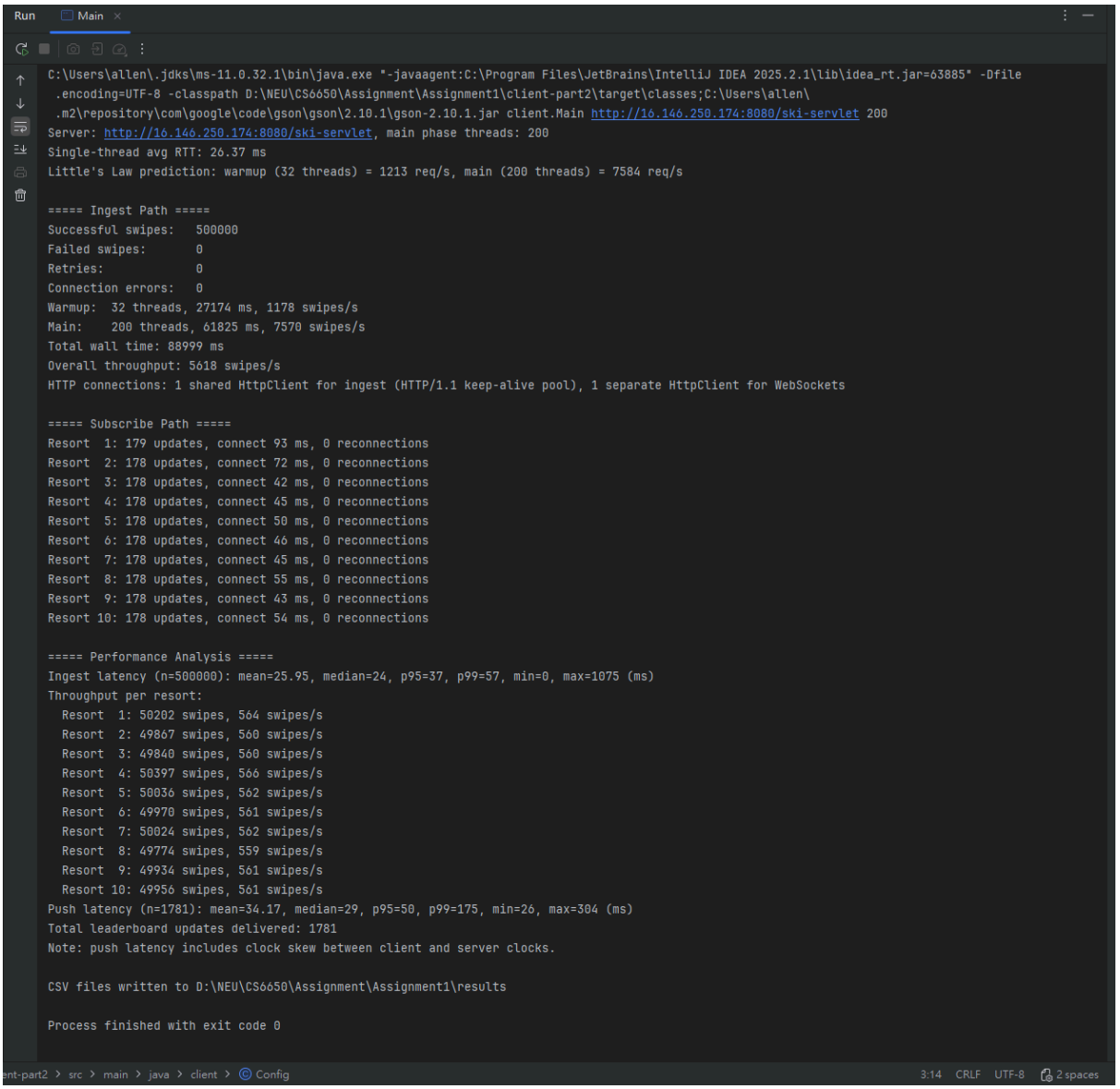
3. Test Results

- Part 1 Output



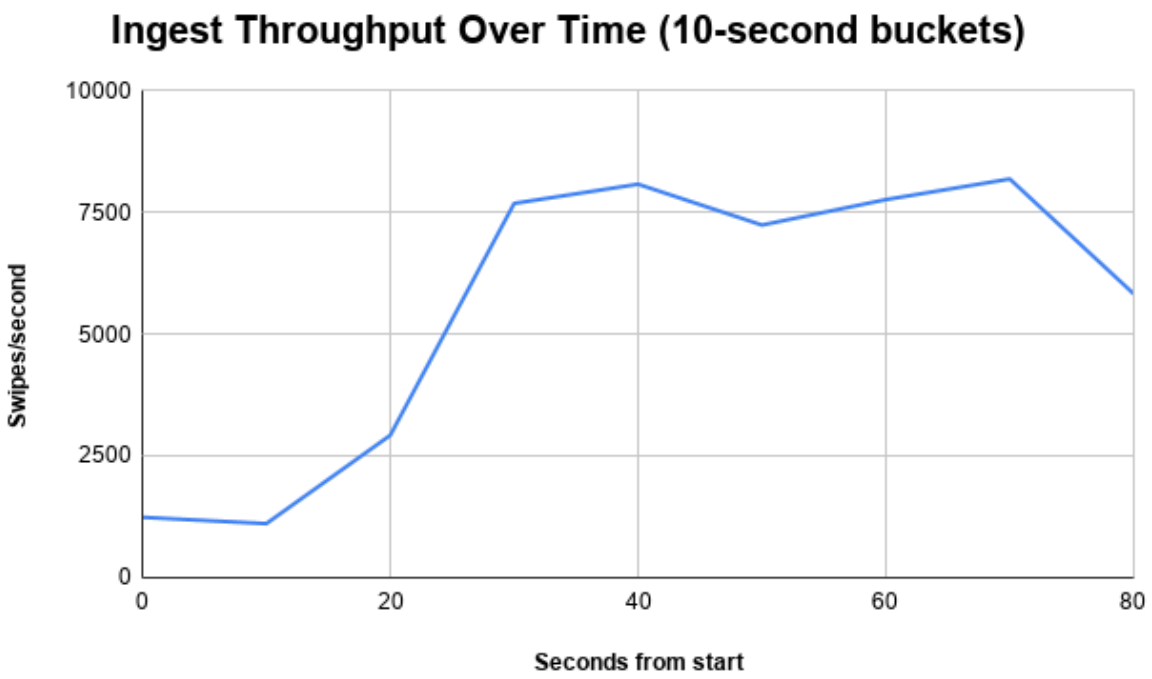
500,000 successful swipes, 0 failed. Warmup 1,239 swipes/s, main phase 6,699 swipes/s, overall 5,225 swipes/s.

- Part 2 Output



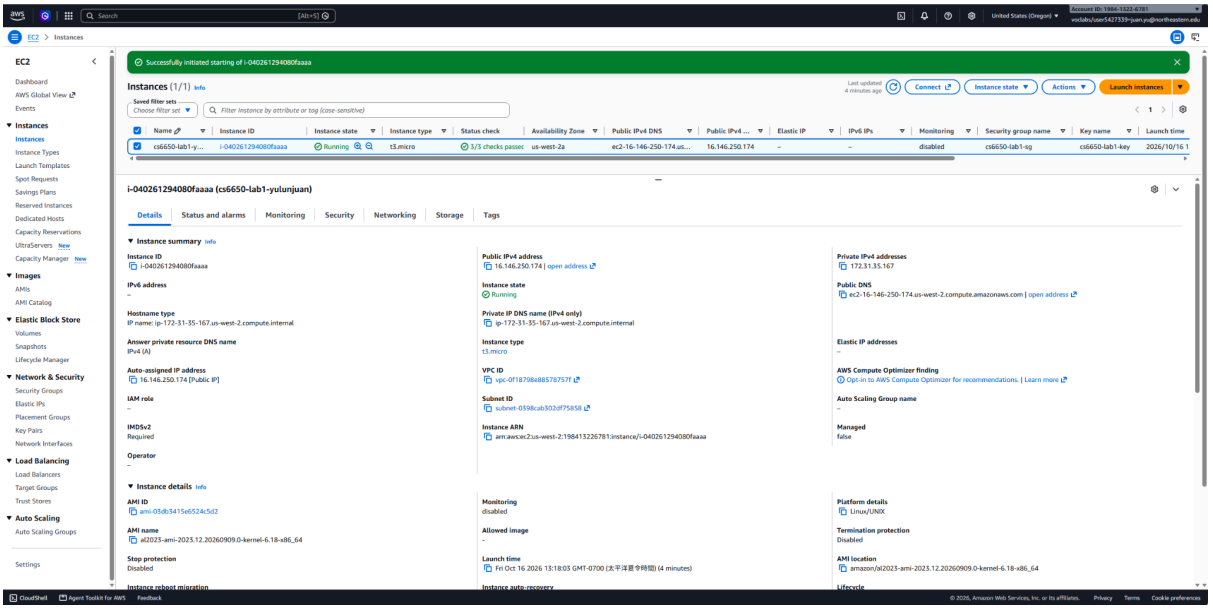
Ingest latency: mean 25.95 ms, median 24 ms, p95 37 ms, p99 57 ms. Push latency: mean 34.17 ms, median 29 ms, p95 50 ms, p99 175 ms. Overall throughput: 5,618 swipes/s. Throughput per resort was about 560 swipes/s for each of the 10 resorts, since resortID is randomly generated with a uniform distribution.

- Throughput Over Time

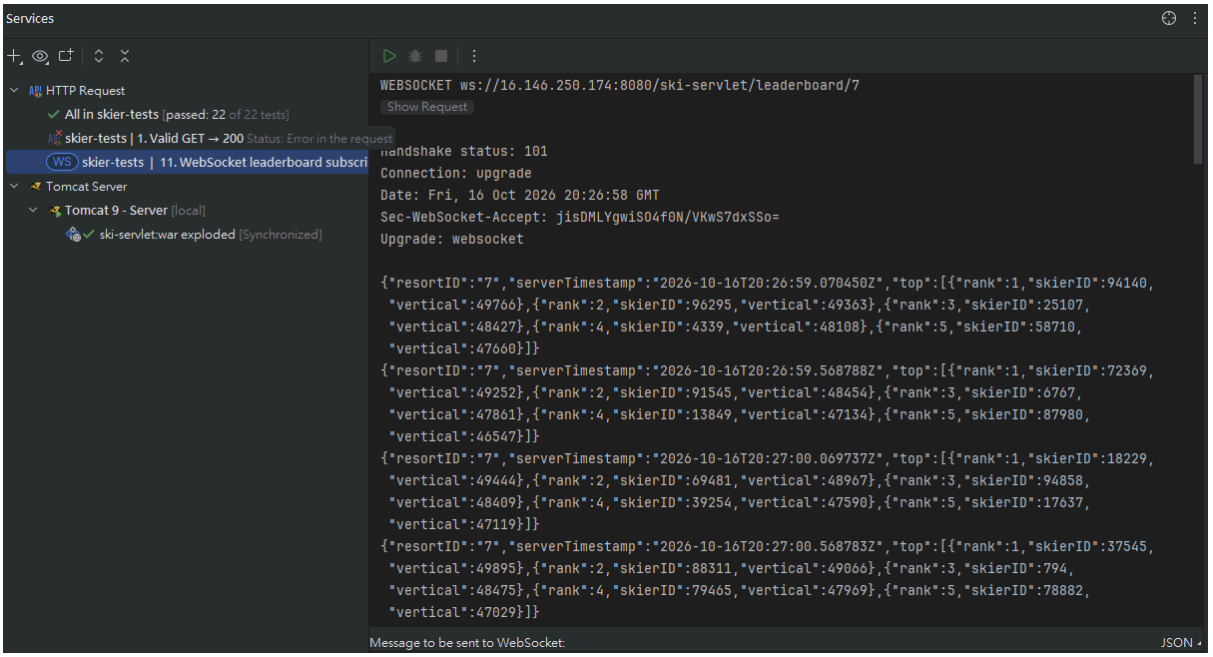


The first 20 seconds are the warm-up phase. The main phase runs at about 7,200–8,200 swipes/s. The last bucket is lower because it covers less than 10 seconds.

- EC2 Deployment



- WebSocket Subscriber Receiving Leaderboard Pushes



4. AI Usage Reflection

See AI_USAGE.md in the repository.